THE Z-PLANE AND THE JURY TEST: INSTABILITY FROM SLOW SAMPLING

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1.0 Learning Outcomes

Upon completing this practical exercise, students should be able to demonstrate the Learning Outcomes presented in Table 1. At the end of the exercise, revisit each Learning Outcome and confirm that it has been achieved.

Table 1: *Learning Outcomes for the Practical Exercise*

ID	Learning Outcome	Achieved
1	Derive the discrete-time characteristic equation of a sampled proportional speed loop with a one-sample delay.	☐
2	Apply the Jury test to find the maximum stable proportional gain as a function of the sample period.	☐
3	Measure the gain limit on the DC motor at three sample periods, and relate the closed-loop pole locations in the z-plane to the measured responses.	☐

2.0 Hardware Set-Up

This practical exercise builds on the hardware and algorithm set-up described in the practical exercise, DC Motor Speed Measurement using an Encoder, which details the operation of a DC motor with an encoder and the measurement of motor speed in revolutions per minute (RPM). The set-up is not repeated here.

The first-order model of the motor, $G(s) = K/(\tau s + 1)$, identified in the practical exercise, Parameter Estimation of a DC Motor using Linear Least Squares, is used throughout. Use the values of K and τ identified for your own motor. Where example values are needed, this guide uses the A1.4 example results, $K = 28$ RPM/V and $\tau = 0.15$ s.

Safety: this exercise deliberately takes the loop to the edge of instability. Keep the output saturation at 0–5 V, use small setpoint steps, and stop the test as soon as a sustained oscillation appears.

3.0 Practical Investigation via Simulink

MATLAB and Simulink should be used to complete this practical exercise. A proportional speed loop that is stable for every gain in continuous time is shown to

become unstable when it is sampled with a delay. The Jury test predicts the gain limit for each sample period, and the predictions are checked on the hardware. The exercise supports the ACE-CORE learning guide, The z-Plane: Stability, Performance and the Jury Test.

Derive the Characteristic Equation

The ZOH model of the motor from the practical exercise, Discrete-Time Modelling of a DC Motor, is $y[k+1] = a\,y[k] + K(1-a)u[k]$, with $a = e^{-T_s/\tau}$. A Unit Delay between the controller and the PWM output represents the time the controller takes to compute, so $u[k] = K_p e[k-1]$. The closed-loop characteristic equation is then second order:

$$P(z) = z^2 - a\,z + K_p K\,(1-a) = 0$$

The Jury conditions for a second-order polynomial $z^2 + a_1 z + a_0$ are $P(1) > 0$, $P(-1) > 0$ and $|a_0| < 1$. The first two hold for any positive gain, and the third gives the limit:

$$0 < K_p K < \frac{1}{1-a} = \frac{1}{1-e^{-T_s/\tau}}$$

At the limit, $|a_0| = 1$ and the closed-loop poles lie on the unit circle at $z = e^{\pm j\theta^*}$, where $\cos\theta = a/2$. The speed then oscillates with a period of $2\pi/\theta$ samples. The limit is plotted against the sample period in Figure 1 for the example motor. It falls steeply as the sample period grows, even though the continuous-time loop has no limit at all.

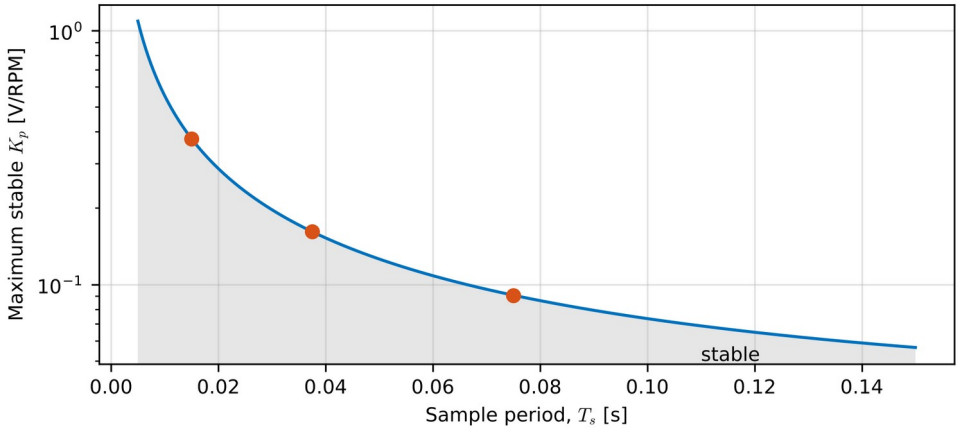


Figure 1: *Theoretical Maximum Stable Proportional Gain against Sample Period, with One Sample of Delay (K = 28 RPM/V, τ = 0.15 s)*

Measure the Gain Limit

[Screenshot to insert] *Step (setpoint 50 → 58 RPM) → Sum (+ −) → Gain Kp → Unit Delay → Sum with 1.8 V feedforward → Saturation 0–5 V → Gain 255/5 → PWM output; encoder speed (RPM) → feedback. All blocks at sample time Ts.*

Figure 2: *Simulink Model for the Sampled Proportional Speed Loop*

1. Build the model in Figure 2. A constant feedforward of 1.8 V is added after the controller so that the motor runs at about 50 RPM, above the dead-zone, before the step.
2. Set all blocks to the sample period T_s = 0.015 s. Apply a setpoint step from 50 RPM to 58 RPM at t = 1 s.
3. Starting at about 30% of the predicted limit in Table 2, raise K_p in steps of about 10% until the speed oscillates without dying away. Record the gain.
4. Repeat for T_s = 0.0375 s and T_s = 0.075 s. Recalculate the limits first with your own K and τ.
5. For one stable gain near the limit at each sample period, plot the closed-loop poles with `pzmap` and `zgrid`.

Table 2: *Predicted and Measured Gain Limits*

T_s (s)	a	Predicted max K_p (V/RPM)	Predicted period (s)	Measured max K_p (V/RPM)	Measured period (s)
0.015	0.905	0.375	0.09		
0.0375	0.779	0.161	0.20		
0.075	0.607	0.091	0.37		

Example responses at T_s = 0.0375 s are shown in Figure 3. They are simulated with the motor model identified in the practical exercise, Parameter Estimation of a DC Motor using Linear Least Squares (K = 28 RPM/V, τ = 0.15 s), so they show the behaviour to expect rather than measured data. Near the limit, the speed oscillates with a period of only a few samples and decays very slowly, the signature of closed-loop poles close to the unit circle.

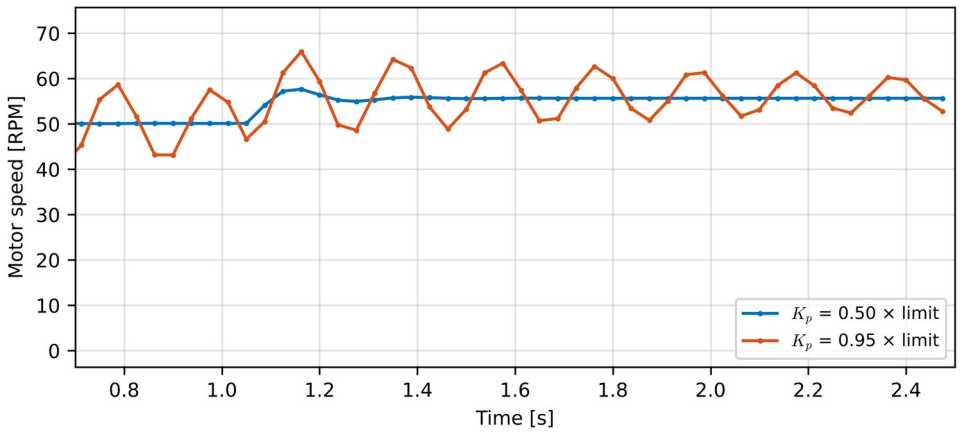


Figure 3: *Simulated Example of the Speed Response at Ts = 0.0375 s for Gains at 50% and 95% of the Jury Limit*

1. Why can sampling and delay destabilise a loop that is stable for every gain in continuous time?
2. Where on the unit circle would the poles at the limit lie for a very long sample period, and what would the response look like?

3. How would you choose the sample period of a controller to keep a comfortable gain margin?

4.0 Concluding Remarks

This practical exercise showed that sampling and computational delay can destabilise a feedback loop that is stable for every gain in continuous time. Using MATLAB, Simulink and ACE-Lab hardware, the Jury test predicted the maximum stable gain of a sampled proportional speed loop, and the predictions were checked at three sample periods.

The gain limit fell sharply as the sample period increased, and the responses near the limit showed the fast, lightly damped oscillation of poles close to the unit circle. Analysing a digital controller in the z-plane, including its delay, is therefore essential whenever the sample period is not very small compared with the dynamics being controlled.

Recommended Citation

If you have used this practical exercise in your teaching, learning or research, please cite it using the reference below, adapting the format as required by your preferred referencing style.

Pickering, J. E. (2026). The z-Plane and the Jury Test: Instability from Slow Sampling. ACE-Lab. Teaching Notes: ACE-Lab Practical Exercises, Version 1.0. Zenodo. https://zenodo.org/records/23725343